

Policy on Medical Gases, Vacuum and Compressed Air

«Hospital Name»

Document control

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Issue No.	01	Review due	One year from implementation
Date created	_____	Date of implementation	_____
Prepared by	Engineering In-Charge Name: _____ Signature: _____		
Reviewed by	Quality Coordinator Name: _____ Signature: _____		
Approved by	Medical Superintendent Name: _____ Signature: _____		

A blank marked _____ must be completed before issue.

Statement of intent

The organisation has a programme for medical gases, vacuum and compressed air.

1. Purpose

This policy sets out how «Hospital Name» runs its medical gases, vacuum and compressed-air programme. It covers:

- Governing the procurement, handling, storage, distribution, use and replenishment of medical gases, compressed air and vacuum
- Maintaining safety measures at every level
- Keeping operational and maintenance plans current
- Providing and testing backup sources

This policy covers medical gases, vacuum and compressed air specifically. Related duties — like patient assessment, clinical care, medication management, patient rights, infection control, quality and safety monitoring, or hospital governance — are covered in the hospital's other policies, not repeated here.

Words marked like this are defaults. A blank marked _____ must be filled before issue.

2. Scope

This policy applies to «Hospital Name» staff involved in facilities, utilities, medical gases, and fire or non-fire emergencies, including:

- Engineering, biomedical, nursing and departmental leaders
- The Engineering In-Charge
- The Medical Superintendent
- Departmental leaders
- The Quality Coordinator

This policy covers medical gases, vacuum and compressed air specifically. Related duties are covered in the hospital's other policies, not repeated here, including:

- Patient assessment
- Clinical care
- Medication management
- Patient rights
- Infection control
- Quality and safety monitoring
- Hospital governance

3. Policy standards

Written guidance governs every stage of medical-gas management at «Hospital Name». Gases are handled, stored and distributed with standardised colour coding, alarms, valve boxes, pin-indexed outlets and automatic changeover. An operational, inspection, testing and maintenance plan follows the manufacturer. Backup sources are in place and tested regularly.

Staff follow written guidance and keep the records listed in the traceability table.

4. Non-negotiable rules

1. Do not procure, store, distribute or use any medical gas without written guidance covering colour coding, signage, handling and replenishment in place for that gas.
2. Do not use a medical-gas outlet or manifold that lacks the required colour coding, alarm, valve box, pin-indexed outlet or automatic changeover to the alternate source.
3. Do not silence a plant-room alarm for a medical-gas system without a documented reason and corrective action.
4. Do not operate piped medical gas, compressed air or vacuum without an implemented operational, inspection, testing and maintenance plan following the manufacturer.
5. Do not operate without a required alternate source (stand-by compressor, stand-by vacuum pump, stand-by manifold or bulk cylinders) for each gas, compressed air and vacuum line in use.
6. Do not count an alternate source as available if it has not been tested at the defined frequency with documented results.
7. Do not bypass the stop-work authority in section 6 when the trigger conditions are met.
8. Staff who see a FMS.6 rule broken report it the same shift to the Engineering In-Charge or the Medical Superintendent.

5. What we do

5.1 Written guidance governs the implementation of procurement, handling, storage, distribution, usage and replenishment of medical gases.

Written guidance governs procurement, handling, storage, distribution, usage and replenishment of all medical gases used here, including colour coding and full/empty signage. HTM 02-01 or the National Fire Protection Association medical-gas handbook is a reference.

The Guidebook names the Explosives Act, Gas Cylinder Rules and Static and Mobile Pressure Vessels (Unfired) Rules. Those duties stay with the named statute; this policy does not paste the Act.

5.2 Medical gases are handled, stored, distributed and used in a safe manner

Medical gases are handled, stored, distributed and used safely: colour-coded cylinders and pipelines; alarms and valve boxes; twenty-four-hour monitoring of plant alarms; pin-indexed outlets; automatic change-over to the alternate source.

A live leak or a silenced required alarm is a stop-work trigger (section 6).

5.3 There is an operational, inspection, testing and maintenance plan for piped medical gas, compressed air and vacuum installation.

There is an operational, inspection, testing and maintenance plan for piped medical gas, compressed air and vacuum, following the manufacturer. Compressed-air purity is checked at a terminal outlet at least once a year, at least one terminal in the operating theatre and one in intensive care if those areas exist.

5.4 Alternate sources for medical gases, vacuum and compressed air are provided for, in case of failure.

Alternate sources for medical gases, vacuum and compressed air are provided for failure (stand-by compressor and vacuum pump; stand-by manifold or bulk cylinders).

No required alternate source is a stop-work trigger (section 6).

5.5 The organisation regularly tests the functioning of these alternate sources.

The organisation tests these alternate sources at a defined frequency (default monthly). Results are documented.

6. Stop-work authority

Do not use a medical-gas outlet or manifold when there is a live leak, a required alarm that has been silenced without a recorded reason, or no required alternate source for that gas, vacuum or compressed air.

Stop-work applies to using that outlet or manifold. Immediate life-saving care uses cylinders or the written alternate source.

The person who stops tells the Engineering In-Charge and the Medical Superintendent the same shift. Refusing an unsafe gas outlet is not a disciplinary matter.

7. Governance and responsibility

Role	Responsibility
Medical Superintendent	Accountable that FMS.6 is resourced and followed.
Engineering In-Charge	Owns day-to-day implementation and records for this standard.
Quality Coordinator	Audits this policy quarterly; holds training acknowledgements.
departmental leaders	Run the department-level duties this standard names.

8. Quality monitoring (RCA → CAPA)

The Quality Coordinator audits this policy quarterly. The audit reviews:

- Records for a sample of this standard's objective elements, checked against the What-we-do steps.
- Documentary evidence is on file for each asterisked objective element in the sample.
- CORE objective elements show no critical gaps in the sample.
- Stop-work events (if any) are logged with outcome.

Root-cause analysis is required when a gap found in this audit remains open beyond 90 days.

This policy is reviewed annually, and sooner after a related facility change, utility failure, equipment recall or fire-plan change.

9. Training and staff acknowledgement

Staff covered by this policy are trained at induction and once a year after that. Training covers the What-we-do steps, non-negotiables and stop-work (if present).

Staff acknowledgement

I have read the Policy on Medical Gases, Vacuum and Compressed Air of «Hospital Name». I will follow the processes described.

Name: _____ Designation: _____

Department / floor: _____ Date: _____

Signature: _____

(One row per staff member. The Quality Coordinator holds signed acknowledgements with the induction record.)

10. Distribution

This policy is distributed to:

- Medical Superintendent
- Engineering In-Charge
- Quality Coordinator
- departmental leaders
- staff covered by FMS.6

11. Abbreviations

Abbreviation	Meaning
CAPA	Corrective and Preventive Action
CORE	Core objective element (NABH)
FMS	Facility Management and Safety (NABH Hospitals 6th Edition chapter)
HCO	Hospital (Full Accreditation programme under NABH Hospitals 6th Edition)
NABH	National Accreditation Board for Hospitals and Healthcare Providers
OE	Objective Element

12. Traceability to NABH HCO Full Accreditation 6th Edition FMS.6

This table is an index. It is not how the policy is organised. An asterisk in the Level column means documentation of the process is required.

OE	Level	Requirement	Where addressed	Responsible
FMS.6.a	Commitment*	Written guidance governs the implementation of procurement, handling, storage, distribution, usage and replenishment of medical gases.	Section 3; 5.1	Engineering In-Charge
FMS.6.b	CORE	Medical gases are handled, stored, distributed and used in a safe manner.	Section 3; 5.2; Section 6 Stop-work	Engineering In-Charge
FMS.6.c	Commitment*	There is an operational,	Section 3; 5.3	Engineering In-Charge

		inspection, testing and maintenance plan for piped medical gas, compressed air and vacuum installation.		
FMS.6.d	CORE	Alternate sources for medical gases, vacuum and compressed air are provided for, in case of failure.	Section 3; 5.4; Section 6 Stop-work	Engineering In-Charge
FMS.6.e	Commitment	The organisation regularly tests the functioning of these alternate sources.	Section 3; 5.5	Engineering In-Charge

13. Required Records / Evidence Checklist

Records the hospital holds under this policy, listed by objective element.

FMS.6.a — Written guidance governs the implementation of procurement, handling, storage, distribution, usage and replenishment of medical gases.

- Written medical-gas procurement, handling, storage, distribution, usage and replenishment guidance, including colour-coding and signage.
- Reference used (for example HTM 02-01 or the NFPA medical-gas handbook).
- Statutory-provision record (Explosives Act, Gas Cylinder Rules, Static and Mobile Pressure Vessels Rules) where those apply to this hospital's gases.

FMS.6.b — Medical gases are handled, stored, distributed and used in a safe manner.

- Colour-coded cylinder and pipeline inventory with alarm and valve-box log.
- Twenty-four-hour plant-alarm monitoring record.
- Pin-indexed outlet and automatic-changeover verification record.

FMS.6.c — There is an operational, inspection, testing and maintenance plan for piped medical gas, compressed air and vacuum installation.

- Documented operational, inspection, testing and maintenance plan for piped medical gas, compressed air and vacuum.

- Compressed-air purity test record, at least annually, at operating-theatre and intensive-care terminal outlets.
- Manufacturer-following maintenance log.

FMS.6.d — Alternate sources for medical gases, vacuum and compressed air are provided for, in case of failure.

- Stand-by compressor/vacuum-pump and stand-by manifold or bulk-cylinder inventory.
- Automatic-changeover test record.
- Alternate-source readiness confirmation log.

FMS.6.e — The organisation regularly tests the functioning of these alternate sources.

- Test log for alternate medical-gas sources at the defined interval (default monthly).
- Manufacturer or written-plan interval reference used.
- Corrective-action record for any failed test.

14. References

- National Accreditation Board for Hospitals and Healthcare Providers (NABH), Accreditation Standards for Hospitals, 6th Edition (January 2025) — Facility Management and Safety, standard FMS.6.
- NABH Guidebook to Accreditation Standards for Hospitals, 6th Edition — FMS.6 interpretations.
- NHS Estates. Health Technical Memorandum (HTM) 02-01 — Medical Gas Pipeline Systems. Referenced as guidance for procurement, handling, storage, distribution and colour-coding of medical gases.
- National Fire Protection Association (NFPA). NFPA 99C — Standard for Gas and Vacuum Systems (Medical Gas and Vacuum Systems Installation Handbook, 2018). Referenced as an alternate guidance source for medical gas and vacuum systems.
- Explosives Act, 1884, and Gas Cylinder Rules, 2016 (as amended), Government of India. Referenced for statutory handling and storage of medical gas cylinders, where applicable to this hospital.
- Static and Mobile Pressure Vessels (Unfired) Rules, 1981 (as amended), Government of India. Referenced for statutory requirements on pressure vessels used for medical gases, where applicable to this hospital.

Disclaimer

This document is a template prepared for the guidance of «Hospital Name» and must be reviewed, adapted and formally approved by «Hospital Name» before use. Every entry marked _____ must be completed before the document is issued.

Several requirements in this document are statutory rather than advisory — in particular those arising under statutory provisions for medical gases as named in NABH FMS.6, including the

Explosives Act, Gas Cylinder Rules and Static and Mobile Pressure Vessels (Unfired) Rules where they apply to this hospital's gases. Statutory requirements change, and State authorities impose additional or stricter conditions. «Hospital Name» is responsible for verifying the current text of any rule cited here and the conditions attached to its own authorisations and licences; this document does not constitute legal advice.

«Hospital Name» remains responsible for verifying that it is current and consistent with the edition of the accreditation standard against which it is being assessed. The clinical and technical content reflects recognised national and international guidance current at the date of preparation.

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